

# JUAN ESTANISLAO HOSZOWSKI

## Backend Software Engineer

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### SUMMARY

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Backend software engineer (Ingeniería en Informática, Universidad de Buenos Aires, Dec 2025) with project experience in REST APIs, microservices, and distributed systems across Python, TypeScript, and C++. Built a microservice with 21 REST endpoints backed by PostgreSQL and Neo4j, a fault-tolerant data pipeline running on 40 Docker containers with RabbitMQ, and the networking layer of a multiplayer game server using a custom binary TCP protocol. Delivered every major project in teams of three to five.

### SKILLS

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**Languages:** C++, Python, TypeScript, Java, Go, SQL

**Frameworks:** FastAPI, Spring Boot, Express.js, React, React Native

**Databases:** PostgreSQL, MongoDB, Neo4j, SQLite

**Infrastructure:** Docker, RabbitMQ, GitHub Actions, CI/CD, Git, Linux, CMake, Render, Firebase

**Concepts:** Distributed Systems, Microservices, Concurrency and Multithreading, TCP networking, REST APIs, System Design, TDD/BDD, Agile/Scrum

**Spoken Languages:** Spanish (Native), English (Advanced), French (Advanced)

### PROJECTS

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#### Fault-Tolerant Data Processing System (Team of 3) | [GitHub](#)

Apr 2025 - Jun 2025

**Stack:** *Python, RabbitMQ, Docker, Pandas, TextBlob*

- Built a distributed pipeline deployed across 40 Docker containers, using custom RabbitMQ-based communication middleware to run five analytical queries over 45,000 movies and 26 million ratings.
- Reduced processing time by 50% (68 to 34 seconds) under three concurrent clients by increasing workers from one to three per stage. Configuration-driven worker counts and topology enabled scaling without code changes.
- Implemented automatic crash recovery with a ring-based health-check protocol: each node checks its successor every second and restarts crashed containers without manual intervention.

#### Microservices Social Platform (Team of 5) | [GitHub](#)

Sep 2024 - Dec 2024

**Stack:** *Python, TypeScript, Java, FastAPI, React Native, PostgreSQL, Neo4j, MongoDB, Docker, Firebase*

- Owned the Twits microservice end to end, implementing 21 REST endpoints for posts, likes, reposts, comments, feeds, trending topics, and search.
- Developed Twits and contributed to authentication within a 12-service backend built in TypeScript, Java, and Python and deployed on Render.
- Used Neo4j as a graph database for Twits and users' social relationships, alongside PostgreSQL for relational data; the platform used MongoDB to store real-time chat messages.

#### Multiplayer Artillery Game, Worms-style (Team of 4) | [GitHub](#)

Oct 2023 - Dec 2023

**Stack:** *C++, Box2D, SDL2, Qt, CMake, YAML*

- Owned the networking layer, implementing a custom binary TCP protocol with 23 operation codes and snapshot-based state synchronization. Built a multithreaded server with a separate 30 Hz fixed-timestep loop for each concurrent match.
- Used server-authoritative Box2D physics to keep game-state calculations on the server, with snapshots communicating state updates to clients.
- Worked with the team to deliver 10 weapons with distinct physics behaviors, 18 maps, support for up to seven players, and a Qt-based map editor.

### EDUCATION

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**Universidad de Buenos Aires (UBA), Facultad de Ingeniería**

*Ingeniería en Informática (Computer Engineering)*

Buenos Aires, Argentina

March 2018 - Dec 2025